ROBERT BAIN

COMPUTER VISION | MACHINE LEARNING | SEMICONDUCTOR ENGINEERING

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ENGINEERING PROFILE

Computer vision and software engineer with an M.S. in Artificial Intelligence and hands-on microfabrication expertise. Built real-time perception for autonomous systems, optimized inference for embedded hardware, and engineered a patent-pending neural recording platform from wafer fabrication through electronics and analysis.

TECHNICAL STRENGTHS

Computer vision: Object detection, image segmentation, image registration, camera calibration, stereo depth, 3D tracking, pose estimation, attention mechanisms, geometric vision

ML and software: Python, C++, PyTorch, TensorFlow, OpenCV, MATLAB, Java, JavaScript, SQL, Linux, Git, ROS, transfer learning, data augmentation

Deployment and optimization: ONNX, TensorRT, CUDA, Jetson AGX Xavier, Coral Edge TPU, model pruning, activation sparsity, embedded inference

Semiconductor and hardware: Photolithography, photomask design, thin-film polymers and metals, flexible circuit design, PCB design, KiCad, Fusion 360, AutoCAD

ENGINEERING AND RESEARCH EXPERIENCE

Graduate Researcher, Neural Engineering and Computer Vision | UC Irvine (2023 to 2026)

- Invented patent-pending flexible neural probes using lithography and thin-film polyimide and metals. Developed scalable wafer fabrication and simplified assembly.
- Brought estimated materials and assembly costs to about $12 per probe, compared with roughly $500 reported for an earlier Nature Communications design. Achieved comparable channel count in mice instead of rats.
- Designed compact acquisition electronics, PCBs, connectors, and packaging. Integrated fabrication, circuits, data acquisition, and software into a working research platform.
- Built GPU-accelerated Python data pipelines and validated three-month recordings. Standardized manufacturing and mentored five undergraduates in assembly, QC, and troubleshooting.

Computer Vision and Machine Learning Research | UC Irvine (2024 to 2026)

- Co-developed pairwise learning to improve image regression and classification from limited labels. Evaluated transfer learning and data augmentation. First-author paper submitted to Machine Learning with Applications.
- Reported state-of-the-art facial age prediction with 3.64-year average error, using a compact model trained from scratch on one consumer GPU.
- Developed interpretable and compute-efficient 3D MRI vision models to predict cognitive traits with limited data and identify informative brain regions.
- Built multi-camera acquisition, calibration, and depth-based 3D pose tracking tools using OpenCV and DeepLabCut for behavioral experiments.

Computer Vision Engineer, Autonomous Racing | Global Formula Racing, Oregon State (2019 to 2021)

- Built real-time cone detection and geometric perception for an autonomous racing car, using homographies and Perspective-n-Point. Prototyped camera-LiDAR co-registration.
- Optimized and deployed vision models with ONNX and TensorRT on NVIDIA Jetson AGX Xavier for a national autonomous racing competition.

ADDITIONAL SOFTWARE AND AI EXPERIENCE

Robotics Researcher | Cognitive Anteater Robotics Lab, UC Irvine (2022 to 2023)

- Developed camera-based localization and ROS multisensor navigation for Toyota HSR and Clearpath Jackal, including camera, LiDAR, IMU and odometry transforms. Used ONNX and TensorRT for HSR inference.
- Worked on CARLsim6, a C++ and CUDA spiking neural network simulator, and developed Git-based continuous integration workflows.

Machine Learning Research, Model Compression | Oregon State University (2020 to 2022)

• Used iterative magnitude pruning to train models with 33x fewer active weights while preserving accuracy. Studied robustness to noisy inputs, activation sparsity, and continual learning. Paper

AI Researcher, DARPA Machine Common Sense | Oregon State (2020 to 2021)

- Led development of Theory-of-Mind agent components for DARPA evaluations of intuitive reasoning and task planning. OSU methods placed first in two of three evaluated categories.

Lead Software Developer | Northwest Media, Eugene, Oregon (2018 to 2020)

- Led development of web software serving approximately 150,000 users and public-sector clients. Supervised two junior developers and modernized 68 interactive training courses from Flash to HTML5.

SELECTED EMBEDDED AI PROJECTS

• Built a Surface Pro 4 system that autonomously played Guitar Hero III at 60 FPS, with CNN inference on a Google Coral Edge TPU, screen capture, controls, and OBS recording. Demo

EDUCATION

University of California, Irvine: M.S., Neurobiology and Behavior, September 2026. GPA: 4.0

Oregon State University: M.S., Artificial Intelligence, 2022. GPA: 3.6

Oregon State University: B.S., Biology, Chemistry minor, 2016

PATENT, SELECTED PUBLICATIONS AND CODE

- U.S. Provisional Patent Application No. 64/130,528 (2026). Capillary-force-assembling, Stiffener-free, High Channel Density Neural Probes. Patent pending.
- **Pairwise Learning Increases Data Efficiency in Deep Neural Networks**. Bain, Rashidi, Chrastil. Submitted to Machine Learning with Applications, Oct 2026. PDF | Code
- **Inferring Continuous Cognitive Traits from Limited Neuroimaging Data Using Pairwise Convolutional Neural Networks**. Rashidi, Bain et al. Manuscript, 2026. PDF
- **ePose: Let's Make EfficientPose More Generally Applicable**. Lally, Bain, Alotaibi. 2021. Paper

Selected code: Multi-camera vision | Visual localization | GPU data processing